

Polymers

Question1

Match the following :

List-I (Use)		List-II (Substance)	
A.	Electrodes in batteries	I.	Polypropylene
B.	Welding of metals	II.	Polyacetylene
C.	Toys	III.	Oxyacetylene

The correct answer is

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Options:

A.

A-III, B-II, C-I

B.

A-II, B-III, C-I

C.

A-II, B-I, C-III

D.

A-I, B-II, C-III

Answer: B

Solution:

- **A. Electrodes in batteries:**

- **Polyacetylene (II)** is a conductive polymer that, when doped, can achieve electrical conductivities similar to metals. This property makes it suitable for use as an electrode material in some types of batteries, particularly in the early development of conducting polymer batteries.
- Polypropylene (I) is used for battery cases, not electrodes.
- Oxyacetylene (III) is a gas mixture for welding.
- Therefore, **A matches II.**

- **B. Welding of metals:**

- **Oxyacetylene (III)** is a mixture of oxygen and acetylene gas, which when ignited, produces a very hot flame (up to 3200°C). This high-temperature flame is widely used for welding, cutting, and heating metals.
- Polypropylene (I) and Polyacetylene (II) are polymers and not directly used for welding metals in this context.
- Therefore, **B matches III.**

- **C. Toys:**

- **Polypropylene (I)** is a versatile thermoplastic polymer known for its durability, flexibility, chemical resistance, and relatively low cost. It is widely used in the manufacturing of various plastic products, including many types of toys.
- Polyacetylene (II) is a specialized conductive polymer, not typically used for common toys.
- Oxyacetylene (III) is a gas mixture, not a material for toys.
- Therefore, **C matches I.**

Combining these matches:

A - II

B - III

C - I

Question2

A polymer sample contains 10 molecules each with molecular mass 5,000 and 5 molecules each with molecular mass 50,000 . The number average molecular mass of the polymer sample is

Options:

A.

$$2 \times 10^4$$

B.

$$3 \times 10^4$$

C.

$$2 \times 10^5$$

D.

$$3 \times 10^5$$

Answer: A

Solution:

Step 1: Find the total mass for each type of molecule

There are 10 molecules, each with a mass of 5,000. So, the total mass for these is:

$$10 \times 5000 = 50000$$

There are 5 molecules, each with a mass of 50,000. So, the total mass for these is:

$$5 \times 50000 = 250000$$

Step 2: Add total masses and total number of molecules

Add the total masses together:

$$50000 + 250000 = 300000$$

Add the total numbers of molecules:

$$10 + 5 = 15$$

Step 3: Calculate the number average molecular mass

Use the formula:

$$M_n = \frac{\sum n_i M_i}{\sum n_i} = \frac{300000}{15} = 20000$$

Question3

Which one of the statements, regarding X is not correct?



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Options:

A.

It is a condensation polymer

B.

It is non-biodegradable

C.

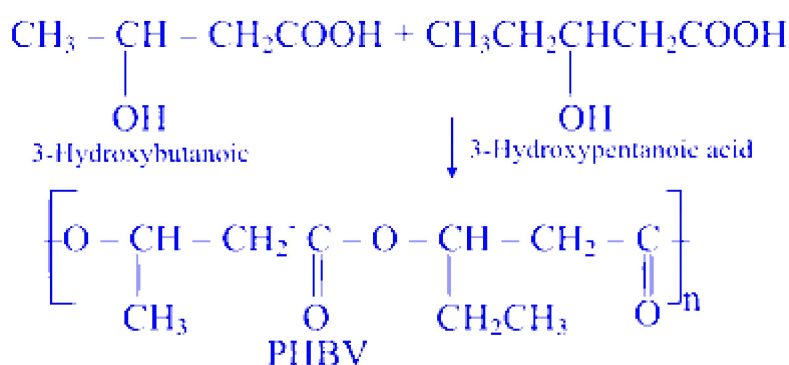
It is used in orthopaedic devices

D.

It is known as PHBV

Answer: B

Solution:



It is a biodegradable polymer.

Question4

Consider the following

Statement I Nylon 6 is a condensation copolymer.

Statement II Nylon 6, 6 is a condensation polymer of adipic acid and tetra-methylene diamine.

The correct answer is

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Options:

A.

Both statement-I and Statement-II are correct.

B.

Statement-I is correct, but Statement-II is not correct.

C.

Statement-I is not correct, but Statement-II is correct.

D.

Both statement-I and Statement-II are not correct.

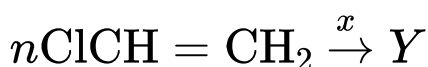
Answer: D

Solution:

Both statement I and statement II are incorrect. This correct forms are Nylon 6 is a condensation homopolymer, Nylon 6,6 is made from adipic acid and hexamethylene diamine.

Question5

What are X and Y in the following reaction?



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Options:

A.

Na/NH₃(l) - Thermosetting polymer

B.

(C₆H₅COO)₂ - Thermoplastic polymer

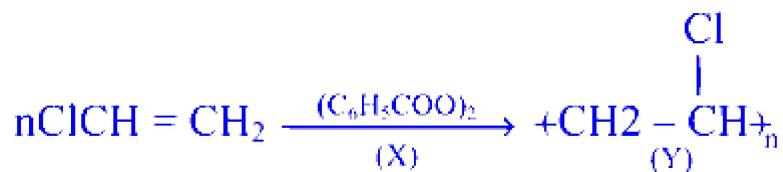
C.

Na/NH₃(l) - Condensation polymer

D.

(C₆H₅COO)₂ - Network polymer**Answer: B****Solution:**

The complete polymerisation reaction is,



Thermoplastic polymer (Product Y).

Question6**Which of the following properties is not correct for silicones?****AP EAPCET 2025 - 23rd May Morning Shift****Options:**

A.

Bio compatible

B.

High thermal stability

C.

Low dielectric strength

D.

Water repelling in nature

Answer: C

Solution:

The incorrect statement about silicon is

It has low dielectric strength.

The correct form is, it has high dielectric strength and due to this it is good insulator.

Question7

Match the following.

List-I (Polymer)	List-II (Used in making)
A. $[-CF_2-CH_2-]_N$	I. Safety helmets
B. $\begin{array}{c} \text{+CH}_2-\text{CH+} \\ \\ \text{CN} \end{array} \text{ }_N$	II. Gaskets
C. $\text{OCH}_2\text{CH}_2\text{O}-\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{C} \text{ }_n$	III. Laminated sheets
D. $\text{NHCONHCH}_2 \text{ }_n$	IV. Commercial fibres
	V. Paints

Correct answer is

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Options:

A.

A-III, B-V, C-I, D-II

B.

A-I, B-II, C-IV, D-III

C.

A-II, B-IV, C-I, D-III

D.

A-II, B-IV, C-V, D-I

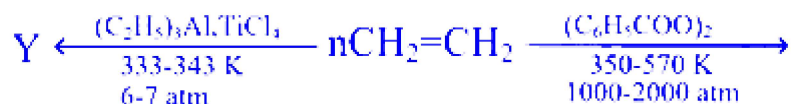
Answer: C

Solution:

The correct match is, A-II, B-IV, C- I, D-III

Question8

The correct statement regarding X and Y in the following set of reactions is



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Options:

A.

X is HDP and Y is LDP.

B.

X is LDP and Y is HDP.

C.

X is used in the preparation of flexible pipes and Y is used in manufacturing squeeze bottles.

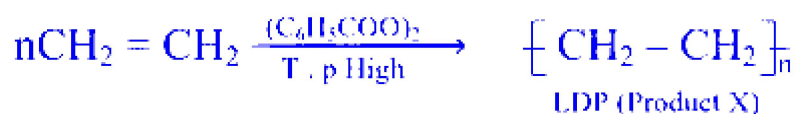
D.

X is used in insulation of electricity carrying wires, Y is used in manufacturing of bottles.

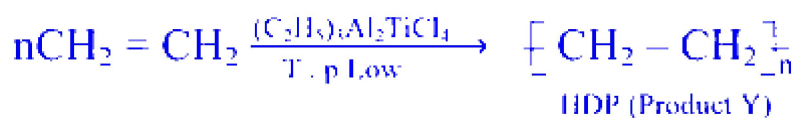
Answer: B

Solution:

Complete reaction is,



Free radical polymerisation



(Ziegler - Natta polymerisation)

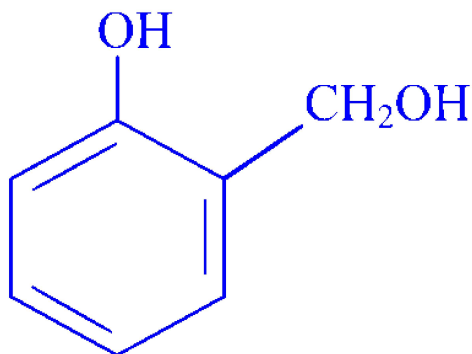
Question9

Novolac is formed by the polymerisation of monomer ' x ' in the presence of OH^- ions. What is ' x '?

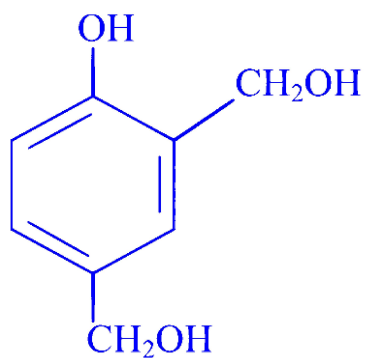
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Options:

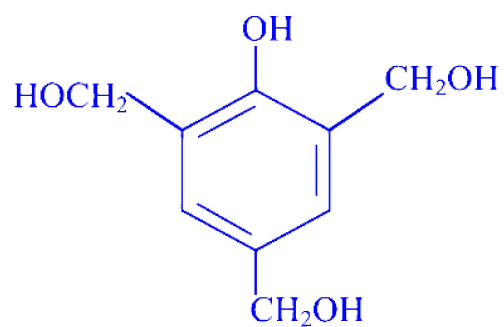
A.



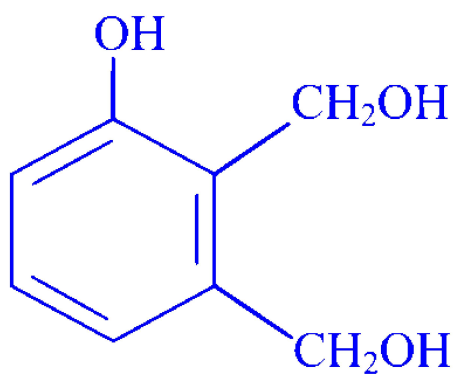
B.



C.



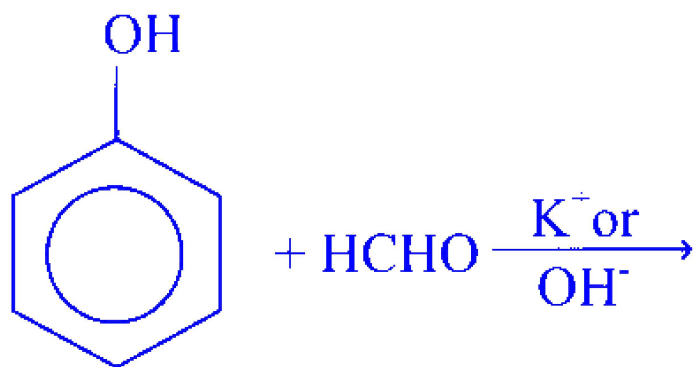
D.

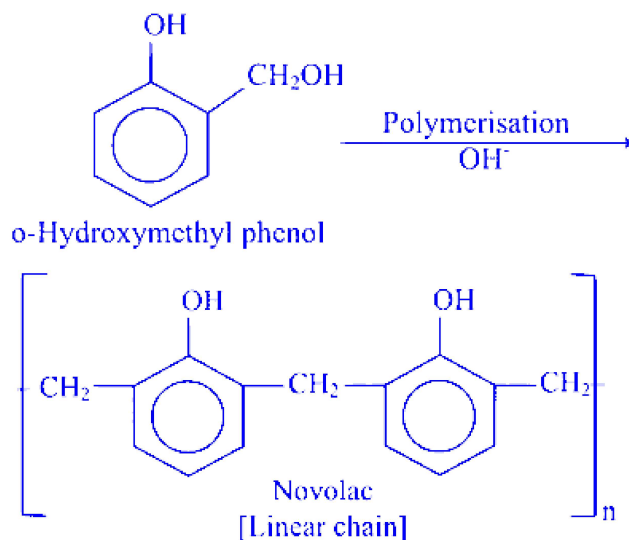


Answer: A

Solution:

The complete polymerisation is,



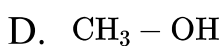
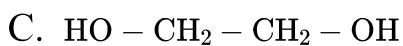
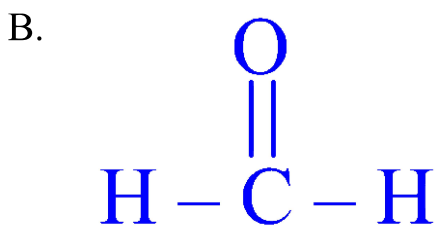
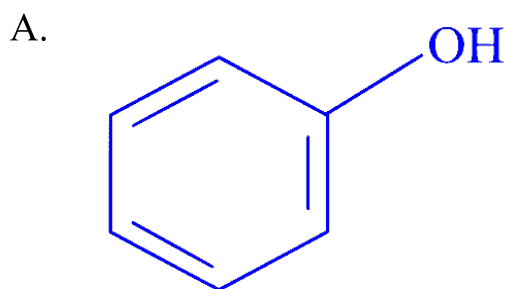


Question10

Which of the following is the common monomer for the polymers bakelite and melamine?

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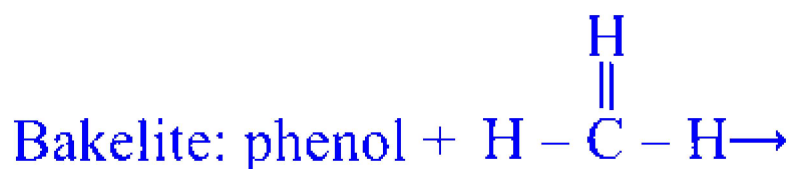
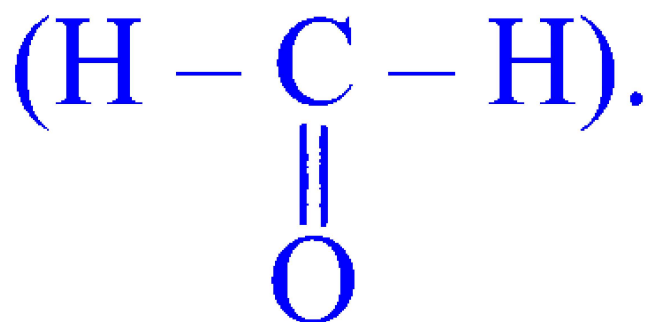
Options:



Answer: B

Solution:

The common monomer for bakelite and melamine is formaldehyde



Phenol-formaldehyde resin



Melamine-formaldehyde resin.

Question11

In which polymer preparation, Ziegler- Natta catalyst is used?

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Options:

A.

Low density polythene

B.

Telfon

C.

Polyacrylonitrile

D.

High density polythene

Answer: D

Solution:

High density polythene is prepared using Ziegler-Natta catalyst. Ziegler - Natta catalyst: A transition metal halides + Organoaluminium compound e.g. $\text{TiCl}_4 + \text{Al}(\text{C}_2\text{H}_5)_3$

Question12

Polymers that can be softened on heating and hardened on cooling are called

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Options:

A. thermosetting polymers

B. bakelite

C. fibres

D. thermoplastic polymers

Answer: D

Solution:

The correct answer is Option D: thermoplastic polymers.

Here's why:

Thermoplastic polymers soften when heated and harden again upon cooling. This property allows them to be remolded repeatedly.

In contrast, thermosetting polymers (such as bakelite) undergo a curing process that permanently locks their structure, so they do not soften upon reheating.

Fibres refer to materials with long, flexible strands and are not classified based on heat-softening properties. Thus, polymers that can be softened on heating and hardened on cooling are called **thermoplastic polymers**.

Question13

Which of the following is correct statement?

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Options:

- A. Starch is a polymer of β -D-glucose
- B. Amylose is a components of starch
- C. Proteins are biopolymers of only one type of amino acids
- D. Lactose is a disaccharide of $\alpha - D$ - glucose and β -D galactose

Answer: B

Solution:

Statement given in option (b) is correct, while all other are incorrect.

The correct statements are:

- (a) Starch is a polymer of $\alpha - D$ glucose.
- (c) Proteins are made of up to 20 different amino acids, so protein is heteropolymers or copolymers.
- (d) Lactose is a disaccharide of $\beta - D$ glucose and β -D-galactose.

Set given in option (a) is not

Question14

Match the following.

List-I (Polymers)		List-II (Type)	
A.	Buna- N -rubber	I	Fibre

List-I (Polymers)		List-II (Type)	
B	Terylene	II	Thermosetting polymer
C	Polystyrene	III	Elastomer
D	Urea-formaldehyde resin	IV	Thermoplastic polymer

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Options:

A. A - III, B - I, C - IV, D - II

B. A - III, B - IV, C - I, D - II

C. A - I, B - II, C - III, D - IV

D. A - IV, B - III, C - I, D - I

Answer: A

Solution:

The correct match is A-III, B-I, C-IV, D-II.

A. Elastomer are any rubber material composed of long chain like polymers that are capable of recovering to original shape after stretching → Buna-N Rubber is elastomer. (Inter-particle force is weakest : van der Waals' force)

B. Fiber forming polymers are linear macromolecules that are suitable for synthetic fiber → Terylene is synthetic fiber.

(Inter-particle force is strongest : H-bonding)

C. A thermoplastic polymer are those polymer that can soften on heating and then processed. Polystyrene is an example.

D. Thermosetting polymers are semifluid substance with low molecular mass, when heated in mould undergo a permanent change, in composition. Urea-formaldehyde resin is an example.

Question15

The correctly matched set of the following is

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Options:

- A. polystyrene - copolymer - thermoplastic
- B. Bakelite - addition polymer - thermosetting
- C. Nylon 6 - homopolymer- fibre
- D. Buna- N - homopolymer - elastomer

Answer: C

Solution:

Option A: Polystyrene is formed by the polymerization of styrene and is a homopolymer (made from a single type of monomer), not a copolymer. Although it is a thermoplastic, the first part of the match is incorrect.

Option B: Bakelite is a thermosetting resin, but it is produced via a condensation reaction (not addition), so calling it an "addition polymer" is incorrect.

Option C: Nylon 6 is produced from caprolactam by ring-opening polymerization, making it a homopolymer. It is widely used in the textile industry as a fiber.

Nylon 6 (a homopolymer of caprolactam) → Fiber

This option is correctly matched.

Option D: Buna-N, also known as nitrile rubber, is a copolymer (usually made from butadiene and acrylonitrile), not a homopolymer, although it is an elastomer.

Thus, the correctly matched set is:

Option C: Nylon 6 – homopolymer – fiber.

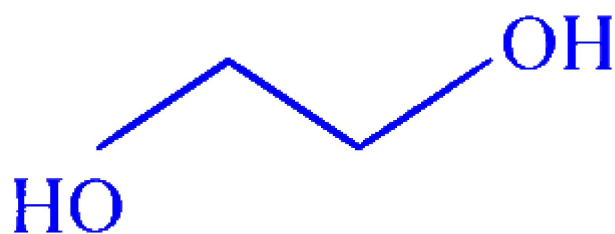
Question16

The common monomer for both terylene and glyptal is

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Options:

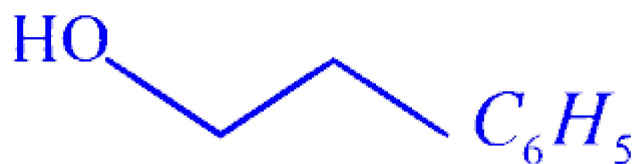
A.



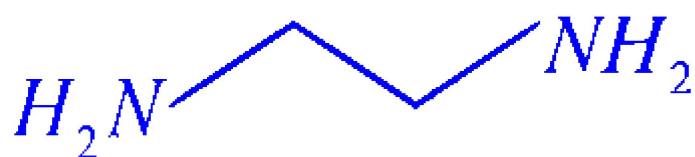
B.



C.



D.

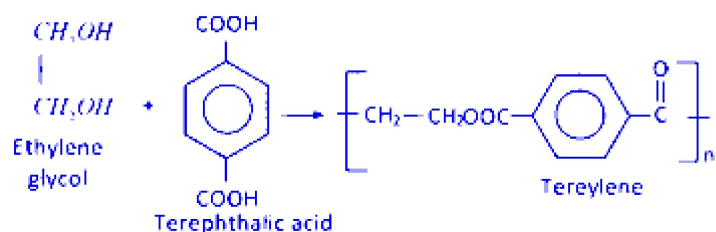
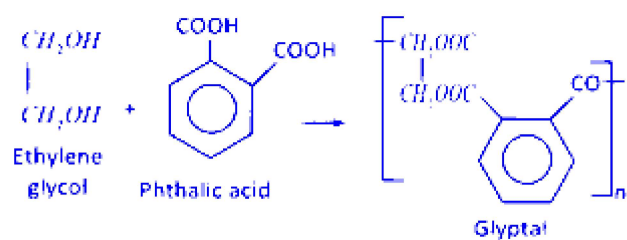


Answer: A

Solution:

Ethylene glycol is the common monomer for both terylene and glyptal.

Ethylene glycol

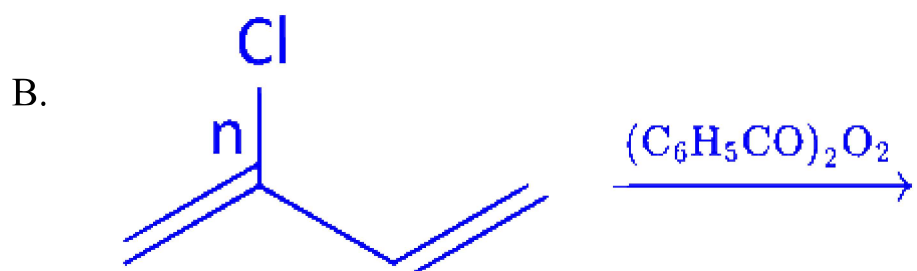
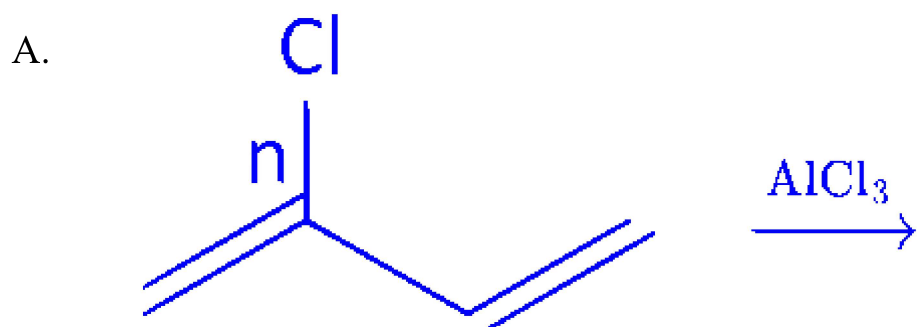


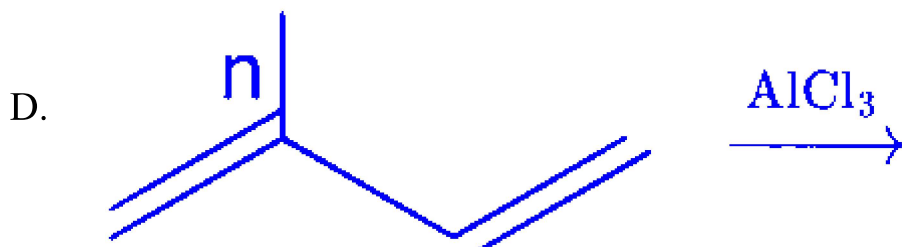
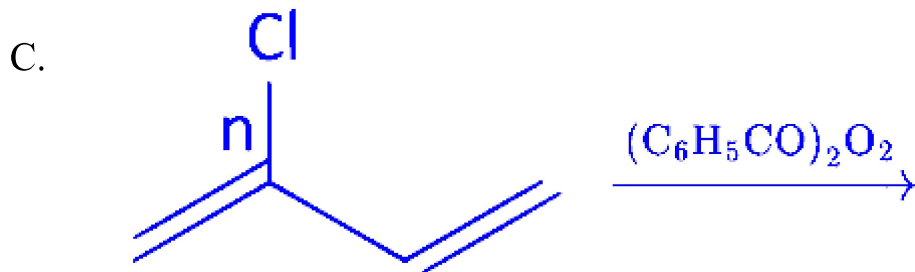
Question17

Which of the following polymerisation leads to the formation of neoprene ?

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Options:

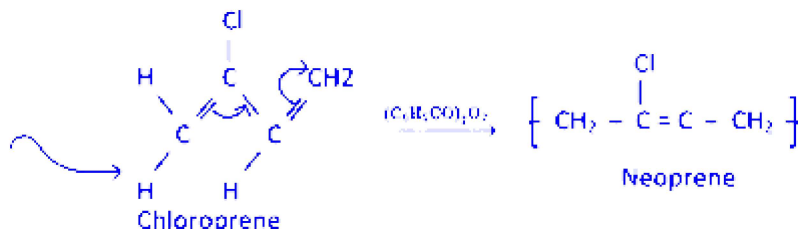




Answer: C

Solution:

Chloroprene is the monomer of neoprene. The complete reaction is as follow.

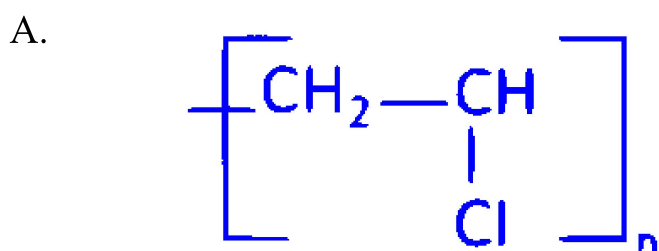


Question18

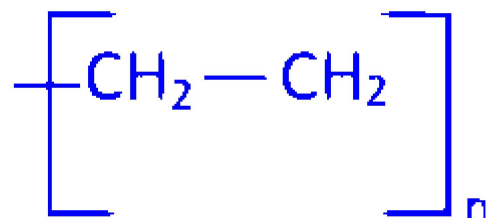
Which of the following is an example for fibre?

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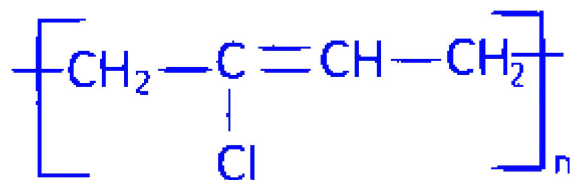
Options:



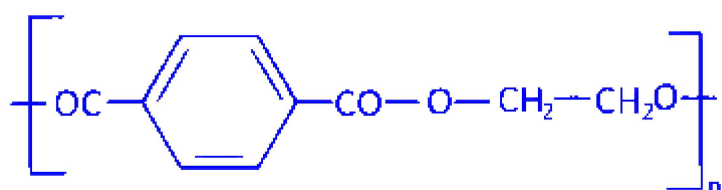
B.



C.



D.

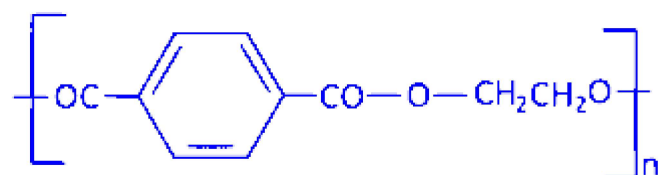
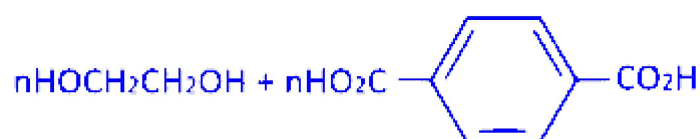


Answer: D

Solution:

Among the given, polyethylene terephthalate is a fibre polymer (a polyester).

Its monomers are ethylene glycol and terephthalic acid.

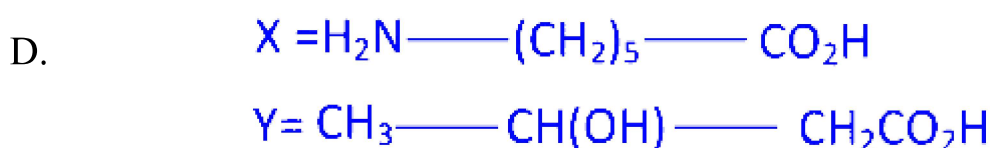
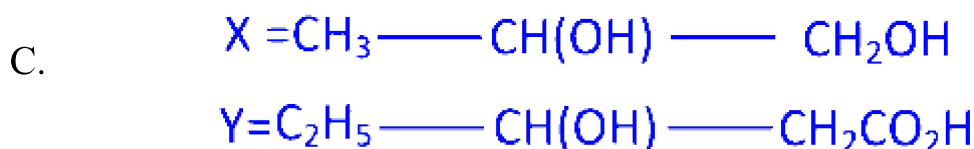
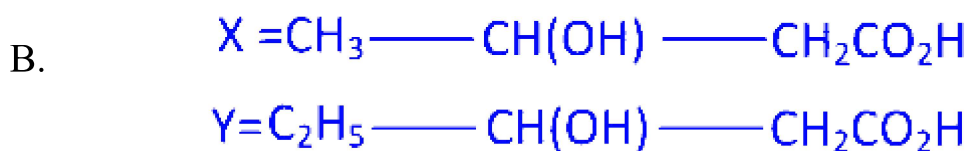
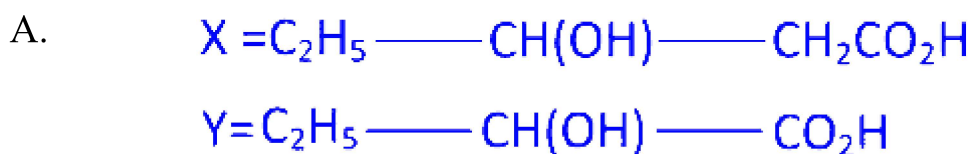


Question19

PHBV is a biodegradable polymer of two monomers X and Y . X and Y respectively are.

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Options:

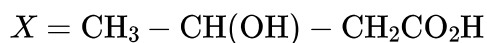


Answer: B

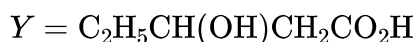
Solution:

PHBV (IUPAC name) 3-hydroxybutyrate-CO- 3 hydroxy valerate) is a polyhydroxy alkanoate type polymer. The monomers are

(i) 3 hydroxybutyrate (3 HB)



(ii) 3 hydroxyvalerate (3 HV)



Question20

A polymer *X* is biodegradable and is obtained from the monomers *Y*, *Z*. What are *Y* and *Z* ?

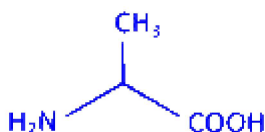
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Options:

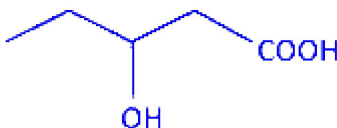
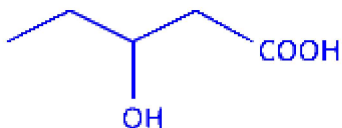
A.



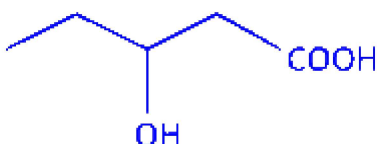
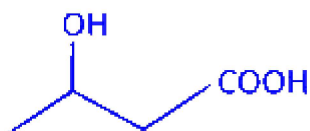
B.



C.



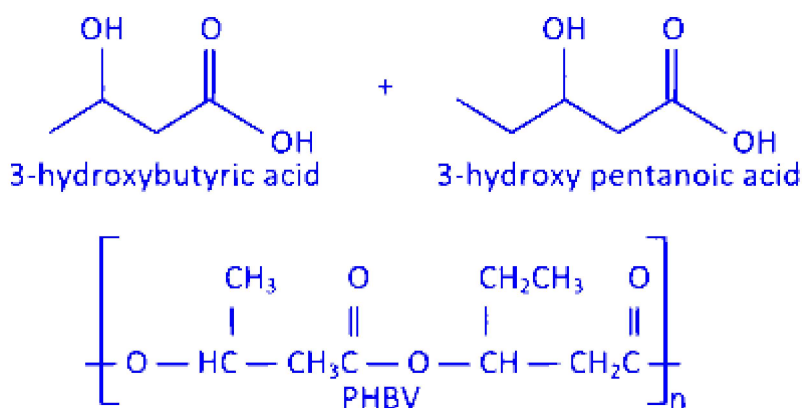
D.



Answer: D

Solution:

Among the given options monomer given in option (d) will give biodegradable polymer i.e., PHBV.



Question21

The catalyst used in the manufacture of polyethylene is a mixture of

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Options:

- A. $\text{Ti}, \text{Al}(\text{CH}_3)_3$
- B. $\text{Ti}, \text{CH}_3\text{MgBr}$
- C. $\text{TiCl}_3, \text{Si}(\text{CH}_3)_4$
- D. $\text{TiCl}_4, \text{Al}(\text{CH}_3)_3$

Answer: D

Solution:

Preparation of polyethylene is done in the presence of the "Zeigler-Natta catalyst" which is a mixture of trimethylaluminium $[\text{Al}(\text{CH}_3)_3]$ and titanium tetrachloride (TiCl_4).